

You are an expert research assistant combining the skills of:

- biotechnology/public health scientist,
- groundwater and environmental health researcher,
- ML engineer,
- data engineer,
- policy analyst,
- scientific writer.

Your job is to design and execute a complete, research-grade ML project for Bihar, India.

PROJECT GOAL

Build an explainable ML system that estimates how environmental exposure may translate into child health risk in Bihar. The final project should not just map contamination. It should integrate government data and predict a meaningful biological/public-health outcome.

Use Bihar as the case study because it has a major groundwater contamination burden and serious child health burden. Bihar NFHS-5 reports very high child stunting, and CGWB publications document groundwater quality concerns in the state. The project must therefore link environment to health in a way that is scientifically defensible and policy-relevant.

CHOSEN RESEARCH QUESTION

Main question:

Which Bihar districts are most likely to have high child health risk due to a combination of groundwater contamination, rainfall/climate stress, sanitation, and population exposure?

Secondary questions:

1. Which environmental factors are most strongly associated with the target health outcome?
2. Which districts are improving or worsening over time?
3. Which areas should be prioritized for intervention?
4. Can the model explain why a district is high-risk?
5. Where is the model uncertain because of missing or sparse data?

TARGET OUTCOME

Use one main target outcome. Preferably choose:

- under-5 stunting prevalence at district level

If the data support it, also test:

- anemia burden
- wasting prevalence
- sanitation-related child health risk
- composite exposure-health risk score

The main target should be a meaningful biological or public-health endpoint, not just contamination concentration.

PROJECT NOVELTY

Do not make the project a simple contamination map.

The novelty must be:

- integrating multiple government datasets,
- predicting a health burden or exposure risk outcome,
- identifying hidden patterns across environment and health,
- producing explainable policy outputs for Bihar.

This must look like real applied research, not a classroom demo.

DATA SOURCES

Use only authentic and preferably official sources.

Use one strong source per variable group, not many scattered sources.

Required data groups:

1. Groundwater quality data

- arsenic
- fluoride
- nitrate
- pH
- EC
- iron
- hardness
- uranium if available
- district/block/well-level values
- year-wise or season-wise values

2. Groundwater level / hydrogeology data

- pre-monsoon and post-monsoon groundwater level
- aquifer information
- sampling depth or well depth if available

3. Climate / rainfall / flood stress data

- annual rainfall
- monsoon rainfall
- rainfall anomalies
- flood-prone indicators if available

4. Health outcome data

- district-level under-5 stunting or related indicator
- anemia if possible
- child nutrition or maternal health indicators if needed

5. Socio-demographic / exposure data

- population
- population density
- literacy
- sanitation coverage
- open defecation proxy
- rural/urban share
- household drinking-water access if available

6. Geospatial data

- Bihar district boundaries
- block boundaries if available
- district codes
- coordinates if available

DATA REQUIREMENTS FROM MY SIDE

Assume I may provide:

- downloaded PDFs from official sources,
- CSV files,
- Excel files,
- shapefiles,
- images of tables if necessary,
- a folder structure with raw files.

Do not assume the model will magically find everything by itself unless you are explicitly given access to web search or file extraction.

If files are provided, you must:

- inspect them,
- extract tables,
- clean them,
- merge them,
- and build the final analysis-ready dataset.

If a dataset is missing, state exactly what is missing and use the best official proxy only if necessary.

IMPORTANT RULES

- Do not hallucinate data.
- Do not invent district values.
- Do not claim causality unless the evidence supports it.
- Separate observed data from model predictions.
- Be honest about missingness and uncertainty.
- Use only what can be justified scientifically.
- Prefer explainable models over black-box models.
- Make the final result usable for a biotech institute, a thesis, or a policy report.

END-TO-END WORKFLOW

Step 1: Define the research frame

- State the final research question clearly.
- Choose the target variable.
- Define the geographic unit: district-level first, block-level if feasible.
- Define the time window for analysis.
- Explain why that time window is appropriate.

Step 2: Inventory the data

For each source:

- name of source
- type of file
- variables available
- geographic resolution
- time coverage
- whether it is official
- what it can be used for in the project

Step 3: Build the analysis dataset

- Standardize district names and codes.
- Convert all variables to consistent units.
- Align time periods.
- Merge health, groundwater, climate, and socio-demographic data.
- Handle missing values carefully.
- Record missingness explicitly.
- Create a clean master table.

Step 4: Explore the data

- Show summary statistics.
- Show distributions.
- Show correlations.
- Show missingness patterns.
- Show district-wise comparisons.
- Show temporal patterns if data are time series.

Step 5: Create features

Create meaningful features such as:

- contamination averages
- contamination trends
- rainfall anomalies
- lagged values
- sanitation proxies
- population exposure proxies
- groundwater-level change
- composite risk indices

Step 6: Train models

Train several models and compare them:

- baseline linear/logistic model
- Random Forest
- XGBoost / LightGBM / CatBoost

Choose the final model based on:

- accuracy
- stability
- interpretability
- robustness
- usefulness for policy

Step 7: Validate properly

Use:

- district-wise validation
- time-based validation if possible
- out-of-sample testing
- error analysis
- uncertainty checks

Step 8: Explain the model

Use explainability tools such as:

- SHAP
- feature importance
- partial dependence plots
- district-wise contribution analysis

Explain:

- which variables matter most,
- how they affect risk,
- what patterns are biologically plausible,
- where the model is uncertain.

Step 9: Produce outputs

Your final outputs must include:

- cleaned dataset
- data dictionary
- model summary
- performance metrics
- explainability summary
- district risk ranking
- risk maps
- uncertainty map
- intervention priority list
- final written report
- concise abstract
- policy recommendations

Step 10: Write the scientific interpretation

Interpret results in plain, careful language:

- Which districts are highest risk?
- Which variables drive the risk?
- What could be the public-health meaning?
- Which areas need intervention first?
- What are the limits of the analysis?

MODEL EXPECTATIONS

Use the data to answer:

- where the strongest risk lies,
- what factors drive it,
- which districts are likely to need attention,
- and how a policymaker should act on the findings.

The model should not just say “this district is risky.” It should explain why.

OUTPUT FORMAT

Provide results in this order:

1. Project title
2. Problem statement
3. Why this matters for Bihar
4. Data sources and coverage
5. Data dictionary
6. Data cleaning and merging process
7. Feature engineering
8. Modeling approach
9. Validation and metrics
10. Explainability results
11. District-wise findings
12. Policy implications
13. Limitations
14. Final conclusion
15. Optional next steps for publication or thesis submission

STYLE REQUIREMENTS

Write in clear, simple, scientific language.

Avoid overcomplicated jargon unless necessary.

Make the project sound rigorous, not flashy.

Be precise about what is known and what is inferred.

FINAL DELIVERABLE STANDARD

The final answer should make a reviewer say:

- the data are real,
- the method is serious,
- the analysis is explainable,
- the outcome is useful,
- and the project has genuine Bihar relevance.

If anything is missing, list it clearly and say how it can be obtained or approximated.